Chicago Housing Risk Analysis: Economic and Housing Structural Determinants of Building Violations

Group Information

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Research Question

1. How do housing violations vary across time and neighborhoods?
2. To what extent are these patterns associated with economic pressure and housing structural features?

Data and Apporach

Data Resources

The building violations dataset is sourced from the City of Chicago Data Portal, contains reported building violations with dates, descriptions & codes, and location information: `building_violations_raw`

The following census resources are drawn from United States Census Bureau, American Community Survey Most recent 5-year release, Cook County, Illinois:
For economic pressure: `median_household_income_B19013.csv` and `rent_burden_B25070.csv`
For housing structural features: `poverty_status_B17001.csv`, `tenure_owner_renter_B25003.csv`, and `units_in_structure_B25024.csv`

Data Progress

- Building Violations (City of Chicago)

1. Clean Data: time window filtering, GEOID Standardization, quality checks, and tract-level structure
2. Violation Rate Construction
   We converted raw violation counts into a standardized rate per 1,000 buildings, using rate = 1,000 × (count / total_buildings), which ensures that higher values indicate more violations relative to local building stock, enabling meaningful spatial and temporal comparisons.

- Census data for economic pressure and housing structural features

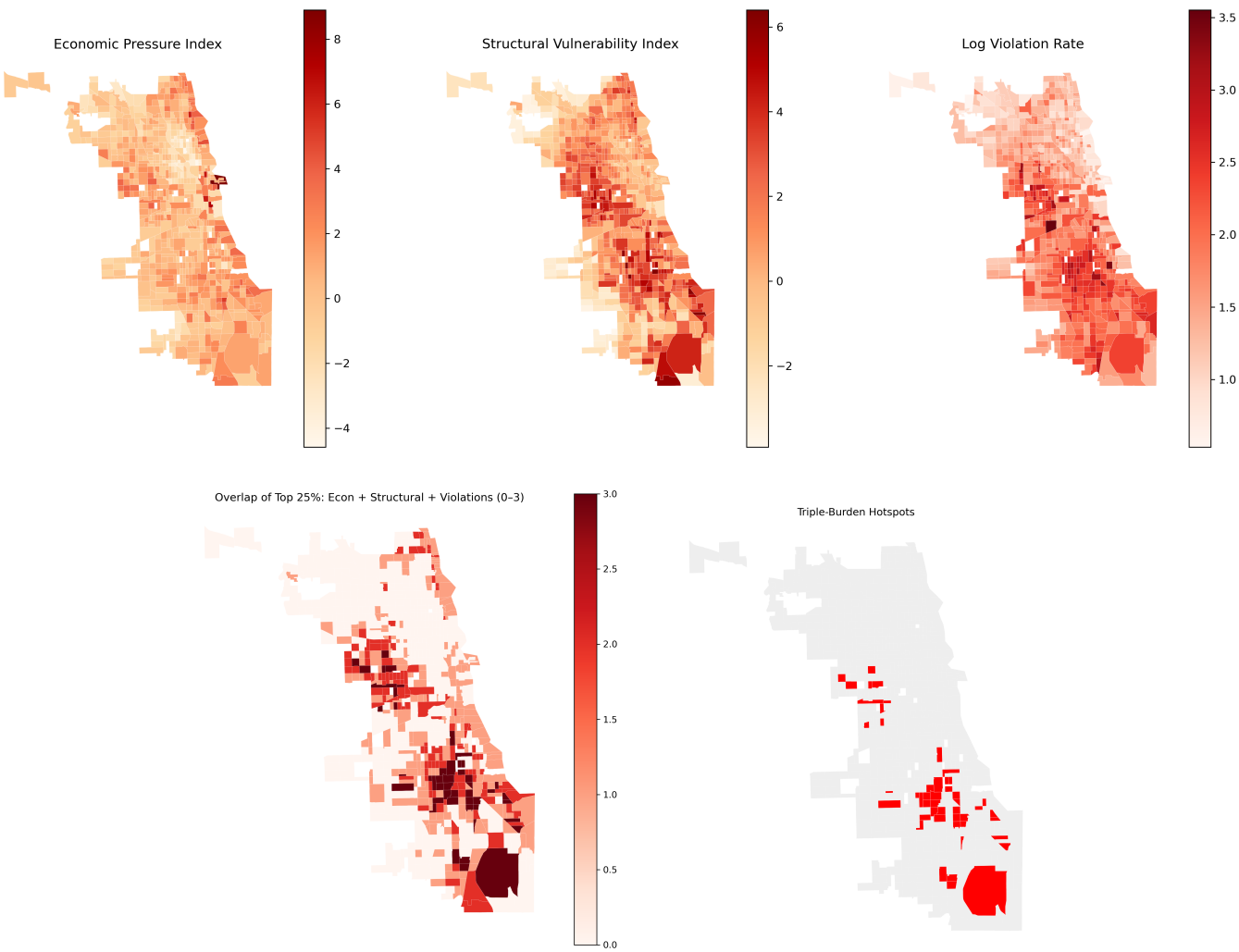
1. Clean Data: GEOID standardization, column & label cleaning, variable selection (estimate only), and missing data handling
2. Standardrization: Economic Pressure Index and Housing Structural Index are created by z-scores Economic pressure Index: rent_burden_z-index (share of households spending 30%) - median_income_z-index (direction reversa)
   Housing Structural Index: poverty_z-index (share of population below the poverty line) + tenure_z-index + housing_age_z-index (share of buildings built in 1939 or earlier)

Weakness and Difficulties

1. Relative rather than absolute measure: Because we use z-score standardization, the index captures relative differences within Cook County rather than absolute levels of housing pressure.
2. Equal wiighting assumption: By aggregating standardized variables, we implicitly assume equal contribution of each component, although their real-world impacts on housing pressure may differ.

Static Visualization

Spatial Distribution



Distribution

Relationship to Research Questions

Correlation Analysis

Distribution

Relationship to Research Questions

Streamlit

Distribution

- Temporal
  A clear break appears in March–April 2020, plausibly reflecting COVID-19 disruptions to inspections after the stay-at-home order. Patterns stabilize from mid-2021, and average rates trend downward through 2024, likely combining real changes and enforcement/recording dynamics.
- Spatial
  Violation rates show persistent within-city imbalance: hotspots repeatedly cluster in Chicago's southwest corridor, while many north-side and lakefront areas remain relatively low. Relative-to-city maps place hotspots in similar locations, suggesting stable spatial concentration rather than random fluctuation.

Relationship to Research Questions

These patterns motivate two questions: Firstly, why do some tracts persistently exhibit higher violation intensity than others; Secondly, to what extent do tract-level economic pressure and structural vulnerability explain the location and persistence of hotspots.